

SOUTIK MANDAL

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📁 PROJECTS

Education

Little Flower School SSC (10th) – 74%	March 2018 – May 2019
Mount Litera Zee School HSC (12th) – 83.4	June 2019 – July 2021
Vellore Institute Of Technology ◦ CGPA: 8.14 ◦ Coursework: Electrical And Electronics Engineering	May 2021 – May 2025

Experience

Electrical Engineer Intern TATA POWER	Jamshedpur, Jharkhand Sept 2023 – Oct 2023
◦ Examined substation distribution systems, transformers, generators ◦ Analyzed thermal power plant generation, working with turbines, boilers, transformers, and DC machines ◦ Collaborated on improvements, and conducted risk assessments to increase turbine efficiency by 90%.	
Electrical and Mechanical Engineer Intern MAFKIN ROBOTICS	Chennai, Tamil Nadu March 2024 – Oct 2024
◦ Contributed to the design of the electrical and mechanical systems for an underwater ship hull cleaning robot ◦ Tested and validated the robot's performance in NIOT National institute of ocean Technology.	
Electrical and Electronics Engineer Intern AEROKNOTZ	Chennai, Tamil Nadu April 2024 – May 2024
◦ Worked on the structural framework and including circuit powering, motor connections, and component integration of an Bluetooth controlled agricultural drone ◦ Developed a battery pack for a power supply that enables up to 6 hours of operation	

Skills

Technical Skills: Robotics Hardware Integration, Electrical Machines, System Design and Automation, Advanced Electrical Engineering, Research and development, Mechanical Systems Design, Actuator Control, Wireless Communication, Safety protocols and risk assessment

Technologies: Matlab Simulink, Proteus, Lt Spice, Auto cad, Easy eda, Tinker CAD, Solid works, C++, Microsoft Excel and PowerPoint

Soft Skills: Technical Writing, Responsive to Evolving Situations, Collaboration & Teamwork, Project Planning Communication, Source checking, Report writing

Projects

Agricultural drone	github.com/projects 🔗
◦ Engineered a reliable electrical system for agricultural drone operations, ensuring crop monitoring and pesticide spraying tasks functioned smoothly across 100+ acres of farmland without operational interruptions during field trials ◦ Tools Used: Pixhawk, AutoCAD, Matlab	

Ship Hull Cleaner Robot

github.com/projects 

- Developed a ship hull cleaner robot to remove rust and debris from vessels, achieving a cleaning efficiency of up to 95% over a 100-square-meter area per hour. Contributed to the mechanical design, enhancing durability and operational ease, and executed testing and validation at the National Institute of Ocean Technology (NIOT).
- Tools Used: Arduino IDE , Proteus

Emergency SOS Vehicle

github.com/projects 

- Designed an Emergency SOS Vehicle for rapid response in critical situations, featuring real-time health monitoring for up to 5 patients, advanced navigation for obstacle clearance within 100 meters, and fast-aid delivery to locations up to 5 to 6 kilometers away

Real-Time Wind Monitoring for Sustainable Energy

github.com/projects 

- Constructed a wind monitoring bot to capture real-time wind speed data and provide alerts during high wind conditions.
- Coordinated with local government and energy experts to secure a 50 lakhs grant for windmill installation, ensuring the project aligned with environmental regulations and promoting sustainable energy use on campus

Automated Agricultural Bot for Precision Farming

github.com/projects 

- Enhanced an agricultural bot that autonomously tills soil, creates rows, and performs seeding, watering, and pesticide spraying, streamlining farming processes for up to 5 acres per day. Integrated multiple sensors for real-time monitoring and automation
- Tools Used: Arduino IDE , Proteus , ThingSpeak

Certifications

1st Place – Agriculture Bot Project

- **Event:** ICDTSA Conference
- Won first prize for the agriculture bot project among 800 participants at the ICDTSA. The bot's autonomous seeding, watering, and pesticide-spraying, along with AI-driven crop monitoring and real-time data analysis, showcased advanced innovation in precision agriculture and automation

2nd Place – Rapid Emergency SOS Vehicle Project

- **Event:** Waveathon, Anna University, Chennai
- Secured second prize among 40 inter college teams at Waveathon, Anna University, for designing an autonomous, solar-powered SOS vehicle. Built for emergency response in highways and disaster zones, the vehicle features autonomous navigation, ECG-based health monitoring, and remote-controlled first aid delivery, recognized for its potential to enhance rapid response and critical aid

Positions of Responsibility

- Directed a team of four interns, providing guidance for robotic projects. Supervised the setup and connections of the SMPS, main power supply, motor, and microcontroller box, ensuring safe and effective integration of all components.
- Student Coordinator and Technical Event coordinator at Cultural Club and TechnoVIT : Organized and managed seven events, including Robo Maze, Robo Trail, and Treasure Hunt, attracting an audience of over 300 attendees. Responsible for event planning, logistics, and on-ground coordination, enhancing participant engagement and ensuring smooth execution.
- Project Lead – Emergency SOS Vehicle Project , Led a team of 4 members to develop a vehicle designed for rapid emergency response. Managed project planning, design implementation, and testing phases, successfully coordinating efforts to secure second place among 40 intercollege teams.